Implications for Computing and Quantum Error Correction

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Author: Quni-Gudzinas, Rowan Brad (QNFO Research Collective) **ORCID:** 0009-0002-4317-5604 **Date:** 2026-08-13 **Version:** v0.3 **License:** CC-BY-4.0 **Keywords:** p-adic valuation; quantum error correction; stabilizer codes; no-cloning theorem; branch depth; ultrametric **Changelog:** v0.3 (2026-08-14) — falsifiability register resolved by derivation: C7 overhead bound DISCONFIRMED (FQ2: valuation data (n,k,q)-only, d inexpressible, bound strictly weaker than Singleton), C5 complexity characterization DISCONFIRMED (FQ3: candidate identities genuine but complexity-vacuous), C6 no-cloning consequence DISCONFIRMED (FQ4: relabels Abramsky's categorical no-cloning; no new QEC consequence). Outcome derivations: artifacts/fq2-overhead-bound.md, artifacts/fq3-complexity-characterization.md, artifacts/fq4-no-cloning-consequence.md. Only C4/FQ1 (non-trivial valuation invariant) remains open. v0.2 (2026-08-13) — §6 precision patch: per-family 83% breakdown, rule-based Algorithm 4.4 reframing, first reproduction attempt documented (Mahler leg not reproduced at n ≤ 18; K–N leg blocked by under-specification), reference corrections (Gubser–Knaute; Bhattacharyya).

Anchor: *Prime Valuation Depth: Multiplication as Branching, the Calculus of Indications, and the Structural No-Cloning Reading* (DOI 10.5281/zenodo.21918838).

Abstract

The anchor paper introduced the *branch-depth reading*: the p-adic valuation $v_p(n)$ is a measure of depth along a prime branch of the integer factorization tree (not a measure of size), and because the tensor product multiplies Hilbert-space dimensions, $v_2(\dim H) = n$ counts tensor-branch depth for n qubits. This follow-on subjects that reading to its own stated falsifiability condition in the two domains the anchor flagged but did not pursue — computing and quantum error correction. The central finding is a *correction, not a confirmation*: the naive mapping of a $[[n, k, d]]$ stabilizer code to branch depths, $n = v_2(\dim H)$ and $k = v_2(\dim H_L)$, is **trivially true by definition** (a code is defined on n qubits encoding 2^k logical states) and therefore carries no new content; and the code distance d — a minimum operator *weight*, not a *depth* — does not admit a valuation reading at all. The genuinely open question is whether there exists a **non-trivial** valuation invariant of stabilizer codes

beyond the definitional n and k, and it is here — not in the naive mapping — that the existing “83% classification accuracy” claim (Kodaira–Néron classifier, DOI 10.5281/zenodo.21193487) lives and must be independently reproduced; a first reproduction attempt is documented in §6 and failed at $n \leq 18$. The structural no-cloning reading is retained but repositioned honestly: it is a known categorical result (Abramsky 2009; Coecke 2009), which the branch-depth vocabulary re-expresses rather than discovers. Every non-empirical claim below carries an explicit falsifiability condition.

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1. Introduction and Positioning

1.1 What the anchor paper established

Prime Valuation Depth developed three claims, each with a stated falsifiability condition:

1. **[TERRITORY — established, Ostrowski 1916]** Every positive integer is a finite product of prime powers; $v_p(n)$ is depth along the prime-p branch, not size.
2. **[MAP — interpretive]** The valuation-as-depth reading is a bridge between the calculus of indications (branching distinctions) and number theory (the prime divisor tree).
3. **[MAP — interpretive]** The tensor product multiplies dimensions, so prime factorization of dim H labels branch types and $v_p(\dim H)$ counts branch depth; the no-cloning theorem is then the impossibility of a linear diagonal map — cloning is nonlinear in the amplitudes, and quantum evolution preserves only linear structure.

1.2 The honest starting point of this paper

The anchor paper labels its own quantum reading as “interpretive rather than new physics” and attaches the falsifiability condition: *disconfirmed if the reading yields no explanatory or predictive content beyond the standard formalism — i.e., if it is pure relabeling.* This follow-on takes that condition seriously and applies it to the two domains the anchor flagged for later work. The result, reported here without flinching, is that **the most obvious extension — reading $[[n, k, d]]$ parameters as valuations — is largely pure relabeling.** This is not a failure of the program; it is the falsifiability machinery doing its job, and it sharpens the question to the one place where the branch-depth vocabulary could carry real weight.

1.3 External precedent (confirmation-bias correction)

The anchor’s quantum reading does not occur in a vacuum. The categorical content — that quantum processes form a compact-closed (monoidal but non-Cartesian) category, and that no-cloning follows from this — is established in the categorical quantum mechanics literature [Abramsky & Coecke 2004; Abramsky 2009; Coecke 2009; Coecke & Duncan 2011]. Independently, the connection between p-adic geometry and quantum error-correcting codes is established in the holographic tensor-network literature [Heydeman, Marcolli, Saberi & Stoica 2018;

Bhattacharyya, Hung, Lei & Li 2018; Gubser & Knaute 2017]. The present paper does not claim priority on either bridge; its narrow contribution is the specific *branch-depth vocabulary* (valuation as depth, in the calculus-of-indications sense) applied to code parameters, and its consequence for how QEC limits are framed.

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2. The Branch-Depth Reading, Restated

Let $n \in \mathbb{Z}^+$ factor as $n = \prod p_i^{a_i}$. The anchor's reading: each prime p_i is a *branch type*, and the exponent $a_i = v_{p_i}(n)$ is the *depth* of nesting along that branch. A tensor product of Hilbert spaces multiplies dimensions: $\dim(H_1 \otimes H_2) = \dim(H_1) \cdot \dim(H_2)$. Hence for n qubits, $\dim H = 2^n$ and

$$v_2(\dim H) = n.$$

This is the single quantitative bridge the anchor supplies. The question this paper asks is: **what does it buy us?**

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3. The [[n,k,d]] Mapping — and Why It Is Mostly Relabeling

3.1 The definitional facts

A $[[n, k, d]]$ stabilizer code is a 2^k-dimensional subspace (the codespace) of an n-qubit Hilbert space $H \cong (\mathbb{C}^2)^{\otimes n}$, with $\dim H = 2^n$. Two identities follow immediately:

$$n = v_2(\dim H) = v_2(2^n) = n,$$

$$k = v_2(\dim H_L) = v_2(2^k) = k.$$

Claim [MAP — to be tested]: these identities constitute a "branch-depth reading" of the code parameters.

Finding [SELF-CORRECTION]: the identities are *definitional*. n is the number of qubits *by construction*, and k is the number of logical qubits *by construction*. Restating them as v_2 of a dimension adds nothing: the valuation is doing no work that the exponent of 2 in "2^n" and "2^k" was not already doing. This is the anchor's own falsifiability condition, triggered: **the n and k mappings are pure relabeling.**

3.2 The distance d does not have a valuation reading

The code distance d is the minimum weight (number of non-identity Pauli factors) of a non-trivial logical operator. It is a **Hamming weight** — a count of tensor factors an error touches — not a p-adic depth. There is no integer whose valuation yields d in the way 2^n yields n. The tempting phrase "d = branch-crossing error weight" is a metaphor, not a valuation identity, and must not be presented as mathematics.

[Falsifiability condition: disconfirmed if any specific valuation identity for d is claimed; d is a weight, and the burden of proof for any valuation-based bound on d is on the claimer.]

3.3 What is left after the relabeling is stripped away

Stripping the definitional n and k mappings and the unavailable d mapping leaves a single, precise, and genuinely open question:

> **Does there exist a non-trivial valuation invariant of a stabilizer code — one not equal to $v_2(\dim H)$ or $v_2(\dim H_L)$ — that carries classification or predictive power across code families?**

This is where the existing internal result attaches.

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4. Computing as Path-Tracing: A Scoped, Falsifiable Claim

The anchor's computing leg was left as a flagged direction. The honest scope is narrow:

Claim [MAP — interpretive, scoped]: reversible classical computation and Clifford quantum computation are *path-tracing through a branching state space*, and the p-adic valuation is the natural depth coordinate on that tree, in the specific sense that the Hensel-code arithmetic of exact rational computation [cf. the p-adic arithmetic literature] is computation *on* the branch tree.

Falsifiability condition: disconfirmed if no computational task can be shown to have a *valuation-based* complexity characterization that differs from (or tightens) its standard characterization. The claim is advanced only as a research program, not as an established result. **[CONTESTED] — no such characterization has yet been produced; this section is explicitly promissory.]**

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5. Structural No-Cloning and the Necessity of QEC

5.1 The honest attribution

The claim "cloning would require a nonlinear diagonal map, and quantum evolution preserves only linear structure" is a known categorical result. In the categorical quantum mechanics framework, a compact-closed category is **non-Cartesian**: the monoidal product ⊗ is not a categorical product, so there is no natural diagonal $\Delta : A \to A \otimes A$ that is linear (a morphism in the category). No-cloning is the physical statement of the absence of this linear diagonal [Abramsky 2009; Coecke 2009]. The anchor paper's contribution is the *vocabulary* — expressing the same fact as "multiplicative branching cannot be linearly duplicated" — not the *result*.

Claim [MAP — re-expression]: the branch-depth vocabulary re-expresses no-cloning as: *a branch (tensor factor) cannot be duplicated by a linear process, so redundancy — the resource QEC consumes — must be carried in non-orthogonal (entangled) configurations, never in clones.*

Falsifiability condition: disconfirmed if this re-expression cannot be shown to yield at least one new, checkable consequence for QEC limits that the standard no-cloning statement does not already imply. **[Risk: HIGH — this is the relabeling risk again, and it is flagged as such.]**

5.2 The one place the framing could bite

A candidate consequence, stated as a falsifiable hypothesis: **if redundancy is non-cloneable, then the overhead (physical-to-logical qubit ratio) of a QEC code is bounded below by a function of the code's own valuation structure.** Whether this bound is tighter than, equivalent to, or weaker than the quantum Singleton bound $n - k \geq 2(d-1)$ is the test. **[RESOLVED — DISCONFIRMED (FQ2, 2026-08-14): the bound has been derived and is strictly weaker than Singleton. The valuation data of a $[[n, k, d]]$ code — $v_2(\dim H)$, $v_2(\dim H_L)$, $v_2(\|S\|)$ — reduce to functions of (n, k, q) only; the Singleton input d is a Hamming weight with no valuation reading (C3, rejected); the sharpest valuation-only bound is $n/k \geq 1$, strictly weaker than Singleton's $n/k \geq n/(n-2(d-1))$ for every code with $d \geq 2$ (378/378 codes in scan). Full derivation: artifacts/fq2-overhead-bound.md.]**

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6. The 83% Classification Claim: Status and Reproduction Requirement

A prior internal report (NTOF, DOI 10.5281/zenodo.21193487) claims that a **rule-based** Kodaira–Néron-fiber classifier (Algorithm 4.4: binary symplectic form H → Cox ring R_C → Weierstrass coefficients → degenerate loci → fiber type) assigns code families with **166/200 (83%)** correct on 50 test codes per family: Surface 46/50 (92%), CSS 39/50 (78%), Optimal 45/50 (90%), Random 36/50 (72%), with Mahler $v_p^{max} = 28$ for optimal versus $v_p^{max} = 4$ for random ensembles. The source itself records a documented partial failure — "FAIL for surface codes (systematic mismatch in the I_n* classification boundaries)" — so the aggregate 83% conceals a known per-family defect. The NTOF record ships no dataset, no implementation, and no baseline; the classifier is deterministic rule-based, not a learned ML classifier. A program-registry summary's phrase "p-adic valuations classify QEC codes at 83% accuracy" is therefore a compressed (and partially misleading) rendering of this state of affairs.

Status [UNVERIFIED-INTERNAL]: this is an internal report. It has not, to this author's knowledge, been independently reproduced on a fresh, held-out test set with a stated leakage-control protocol. Until it is, the 83% figure is a claim to be tested, not a result to be built upon.

First reproduction attempt (2026-08-13, this pipeline's P4.2): the Mahler spectral leg ($C7.3'$) was independently implemented and run on 55 verified codes (CSS $[[7,1,3]]$, $[[15,7,3]]$; toric surface L=2,3; optimal $[[5,1,3]]$; 50 random $[[10,4]]$).

The claimed separation (v_p^{max} optimal ~ 28 vs random ~ 4, gap >= 10) did **not** reproduce at $n \leq 18$ under the weight-enumerator Mahler normalization: observed optimal 4, random median 3 (max 6, which exceeds optimal). Two source under-specifications block a decisive test: the Mahler target function is undefined in NTOF, and Algorithm 4.4's Cox-ring ideal I_C is unspecified. See the reproduction report in the companion artifacts. C8 remains [UNVERIFIED-INTERNAL] pending source clarification.

Reproduction requirement (this paper's P4 critical path): because the classifier is rule-based and the source ships no implementation, the reproduction is a **re-implementation from specification** (Algorithm 4.4) plus **fresh code-family generation** (50 per family; surface, CSS, optimal, random) with (i) a stated generation protocol and seeds, (ii) an independent computation of the v_p-spectral invariant, (iii) baselines (majority-class and random-assignment) reported alongside, and (iv) the source's documented surface-code boundary defect tested explicitly. Full protocol: companion artifacts. Success is reproduction *within stated confidence* (pre-registered acceptance criteria); failure is reported honestly either way.

Falsifiability condition: the "83% accuracy" claim is disconfirmed if independent reproduction fails to exceed a stated baseline by a pre-registered margin. **[Status: first reproduction attempt executed (P4.2, 2026-08-13) did not reproduce the claimed spectral separation at $n \leq 18$; the decisive test remains blocked by source under-specification. This is disclosed, not hidden.]**

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7. Complete Falsifiability Register

#	Claim	Type	Falsifiability condition	Current status
C1	$v_p(n)$ is depth along a prime branch	established (Ostrowski)	—	established
C2	$n = v_2(\dim H)$, $k = v_2(\dim H_L)$ is a "branch-depth reading" of $[[n, k, d]]$	MAP	disconfirmed (pure relabeling)	**SELF-CORRECTED**
C3	d admits a valuation reading	MAP	no valuation identity exists	**REJECTED**
C4	a non-trivial valuation invariant of codes exists	open question	no such invariant found ⇒ program fails	OPEN

#	Claim	Type	Falsifiability condition	Current status
C5	computing = path-tracing, with valuation-based complexity content	MAP	no valuation-based complexity characterization produced	**DISCONFIRMED** (FQ3, 2026-08-14 — candidate identities genuine but complexity-vacuous; artifacts/fq3-complexity-characterization.md)
C6	no-cloning re-expressed as non-cloneable redundancy	re-expression	no new checkable consequence ⇒ relabeling	**DISCONFIRMED** (FQ4, 2026-08-14 — relabels Abramsky's categorical no-cloning; no new QEC consequence; artifacts/fq4-no-cloning-consequence.md)
C7	overhead bounded by valuation structure	hypothesis	compare to Singleton bound	**DISCONFIRMED** (FQ2, 2026-08-14 — valuation data (n,k,q)-only, d inexpressible, bound strictly weaker than Singleton; artifacts/fq2-overhead-bound.md)
C8	Kodaira–Néron classifier is 83% accurate	empirical (internal)	independent reproduction vs baseline; first attempt (P4.2) not reproduced at $n \leq 18$	UNVERIFIED-INTERNAL

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8. Relation to Prior Work (External)

- **Categorical QM / no-cloning:** Abramsky & Coecke (2004) *A categorical semantics of quantum protocols*; Abramsky (2009) *No-Cloning in Categorical Quantum Mechanics*; Coecke (2009) *Quantum Pictorialism*; Coecke & Duncan (2011) *Interacting Quantum Observables*. The no-cloning content of §5 is theirs; this paper adds only vocabulary.
- **p-adic holographic QEC:** Heydeman, Marcolli, Saberi & Stoica (2018) *Tensor networks, p-adic fields, and algebraic curves* (ATMP 22:93); Bhattacharyya, Hung, Lei & Li (2018) *Tensor network and (p-adic) AdS/CFT* (JHEP 2018(1):139, DOI 10.1007/jhep01(2018)139); Gubser & Knaute (2017) *A p-adic version of AdS/CFT* (ATMP 21(7):1655–1683, DOI

10.4310/atmp.2017.v21.n7.a3). The p-adic/QEC connection is theirs; this paper's branch-depth framing is a distinct, narrower lens.

- **Stabilizer methods:** Bravyi, Browne, Calpin, Campbell, Gosset & Howard (2019) *Simulation of quantum circuits by low-rank stabilizer decompositions* — the working background for §3–§4.

Relation to the internal corpus: the p-adic QEC space is densely worked internally (see the due-diligence appendix for the DOI list). This paper's narrow differentiation is the branch-depth vocabulary and the self-correction of §3, which the existing corpus (which treats valuation as a *classifier weight*, not a *depth*) does not perform.

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9. Conclusion

The branch-depth reading, applied honestly to computing and quantum error correction, yields a **negative result and a sharpened open question**. The negative result: the obvious extension — reading $[[n, k, d]]$ parameters as valuations — is definitional for n and k and unavailable for d; it is, in the anchor's own terms, pure relabeling, and this paper says so rather than papering over it. The three falsifiable consequences of the reading have now been derived and each is **disconfirmed**: the valuation-based overhead bound is strictly weaker than the quantum Singleton bound (FQ2), the valuation-based complexity characterization is complexity-vacuous (FQ3), and the no-cloning re-expression yields no new checkable consequence beyond the standard categorical result (FQ4). What remains is the single precise open question — *does a non-trivial valuation invariant of stabilizer codes exist?* — where the 83% classification claim (Kodaira–Néron classifier, DOI 10.5281/zenodo.21193487) is the testable hypothesis at the center. The first reproduction attempt documented in §6 failed at $n \leq 18$ and remains blocked by source under-specification, so the claim must still be independently reproduced before it is treated as established. This is the honest state of the program: a family of falsifiable claims tested, a sharp question remaining, and a prior claim awaiting verification.

Declarations

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Conflicts of Interest: The author declares no conflict of interest.

Data Availability: Reproduction scripts, raw results, and the reproduction report are included as companion artifacts of this record (rq3-mahler-reproduction.py, rq3-results.json, rq3-reproduction-report.md, rq3-reproduction-protocol.md).

Code Availability: The Mahler spectral reproduction implementation is available in the project repository under notebooks/rq3-mahler-reproduction.py.

Author Contributions: Sole author.

Ethics Approval: Not applicable (no human or animal subjects; mathematical and philosophical research).

Consent to Participate: Not applicable.

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